

Computer Networking

Data Communication

❖ Data Communication

- Data means - Data is a raw facts that are collected while information refers to processed data that enables us to take decisions.
- Communication means - exchange of information.
- The process by which data, instructions and information (text,number,images,audio,video) reach from one computer to other computers(wire/wireless) is called data communication.
- In data communication, digital or analog data is transferred between two or more computers, which are connected to each other through a communication channel.
- Data is transmitted from one place to another in the form of signals.

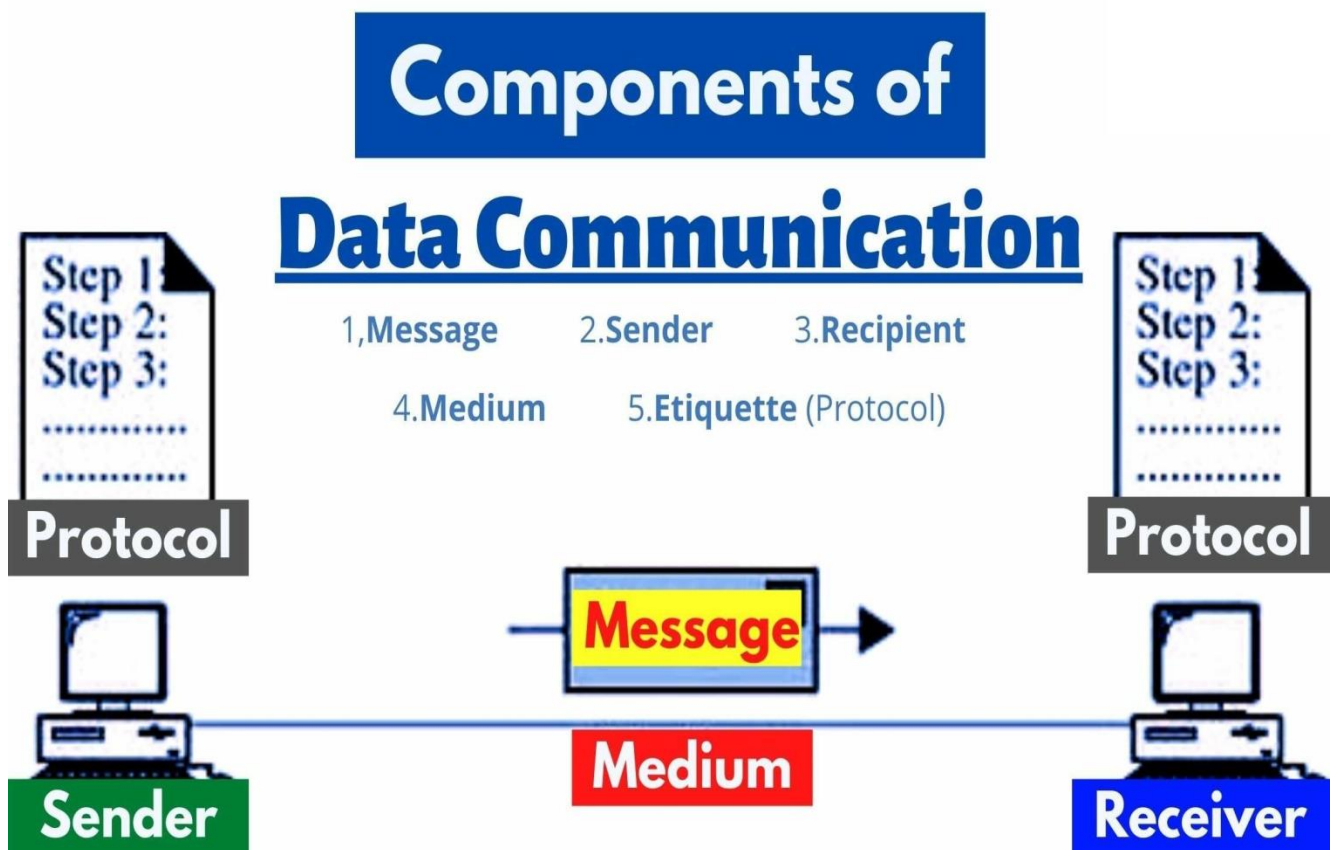
Characteristics of Data Communication

- 1. Delivery:** The data should be delivered to the correct destination and correct user.
- 2. Accuracy:** The communication system should deliver the data accurately, without introducing any errors. The data may get corrupted during transmission affecting the accuracy of the delivered data.
- 3. Timeliness:** Audio and Video data has to be delivered in a timely manner without any delay; such a data delivery is called real time transmission of data.
- 4. Jitter(घबराना):** It is the variation in the packet arrival time. Uneven Jitter may affect the timeliness of data being

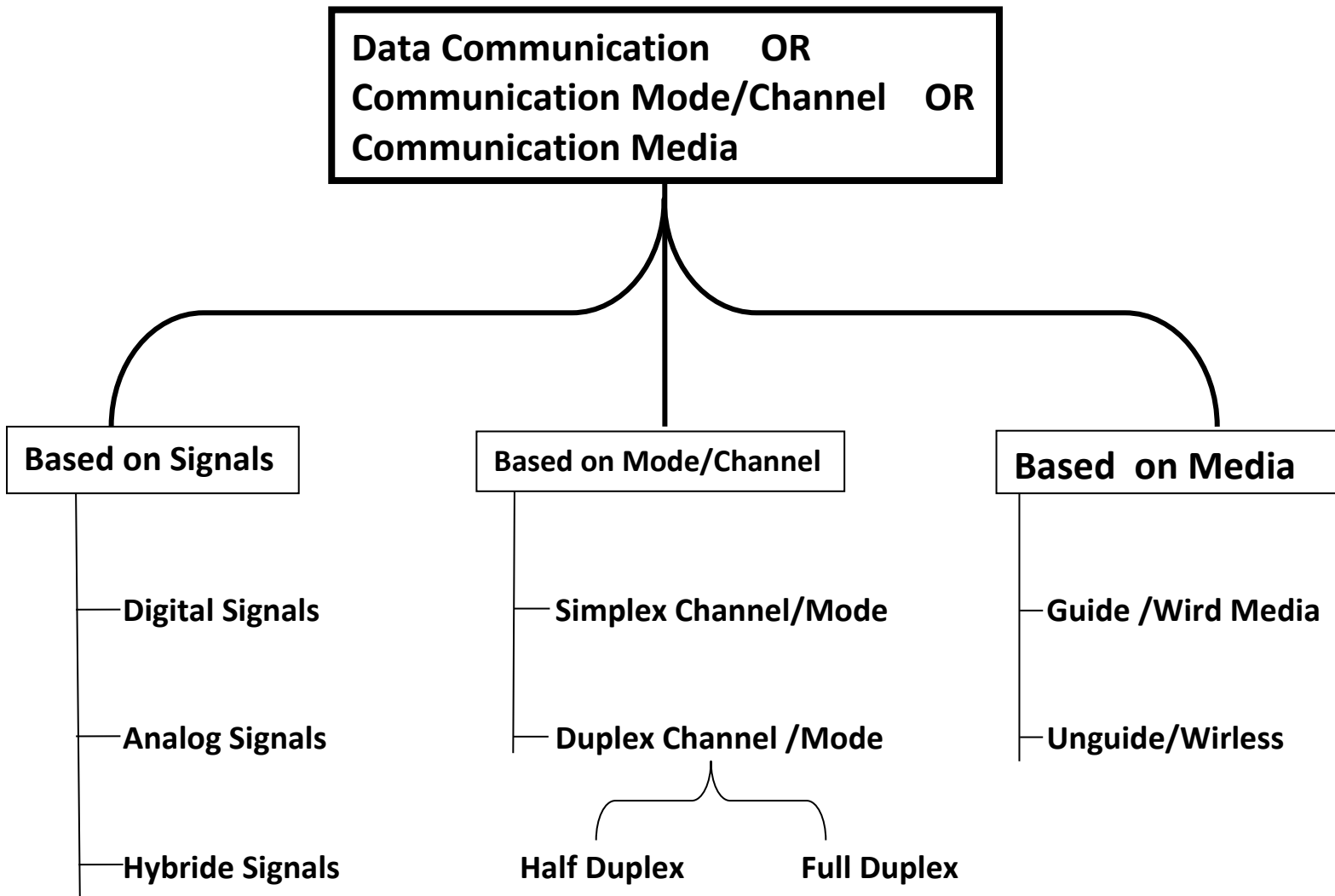
transmitted.

Components of Data Communication

- 1. Message :-** Message is the information to be communicated by the sender to the receiver.
- 2. Sender :-** The sender is any device that is capable of sending the data (message).
- 3. Receiver :-** The receiver is a device that the sender wants to communicate the data (message).
- 4. Transmission Medium :-** It is the path by which the message travels from sender to receiver. It can be wired or wireless and many subtypes in both.
- 5. Protocol :-** A protocol is a set of rules that data communication. It is an agreed upon set or rules used by the sender and receiver to communicate data.



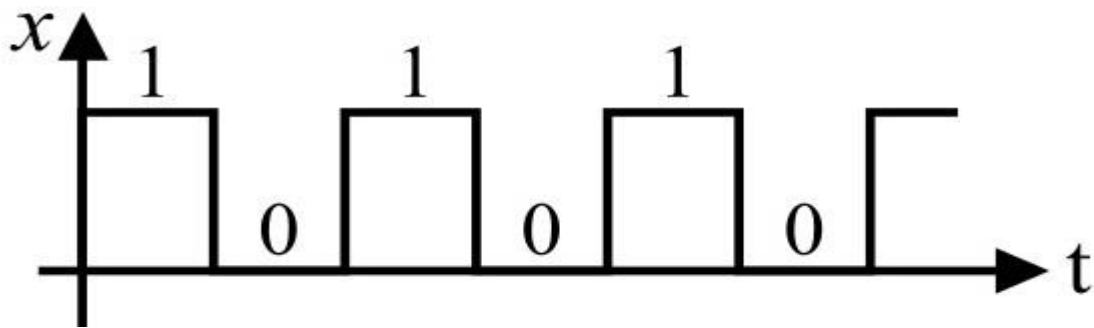
❖ Types of Data Communication



❖ There are three types of signals

1. Digital Signals

- In digital signals, data is exchanged in electronic form, that is, as binary numbers (0 and 1).
- Example - Desktop computers, laptop computers, mobile phones, tablet computers, e-readers, storage devices, such as flash drives, input devices, such as keyboards, mice, and scanners, output devices such as printers and speakers.



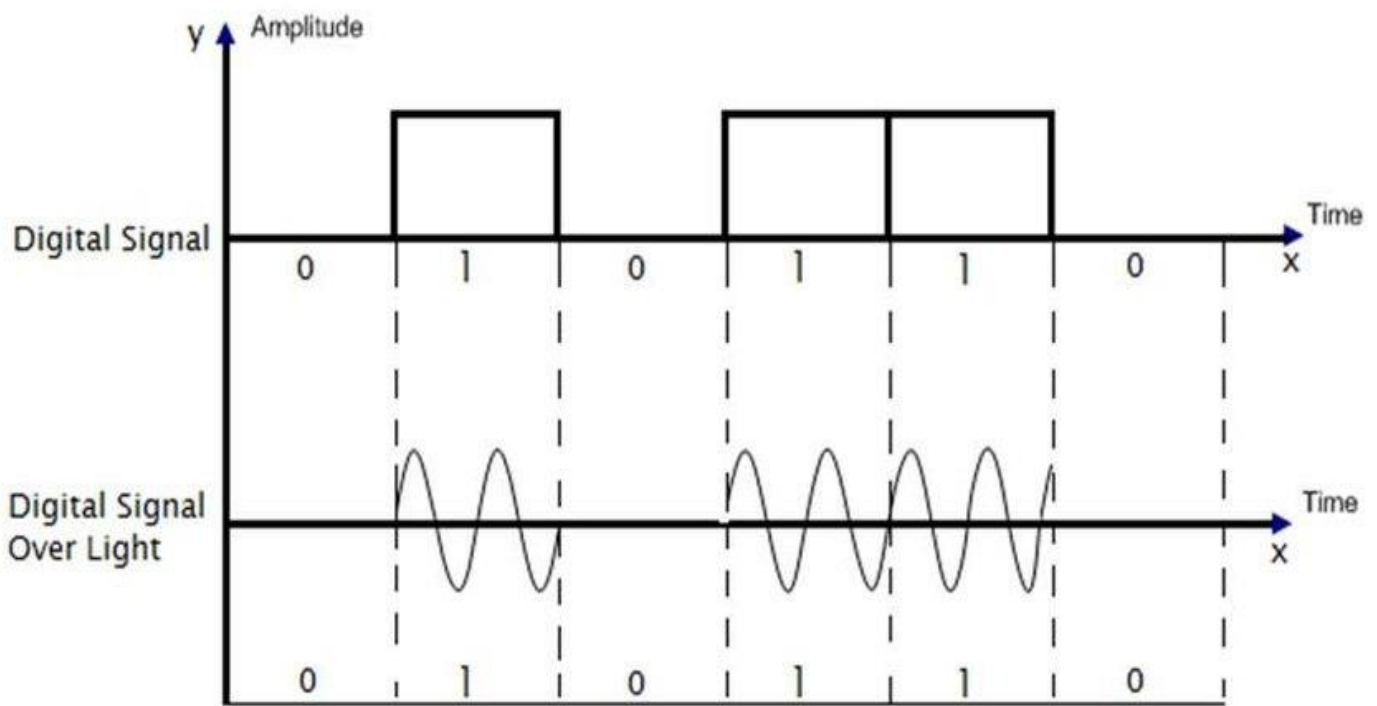
2. Analog Signals

- In analog signals, data is exchanged in the form of radio waves / waves form.
- Example - cassette tapes, records, television through an aerial(हवाई) antenna, basic telephones, and most musical instruments were all analog.



3. Hybrid Signals

- Hybrid signals have properties of both analog and digital signals.
- E.g. – PCM (pulse code modulation)



Difference Between Analog & Digital Signals

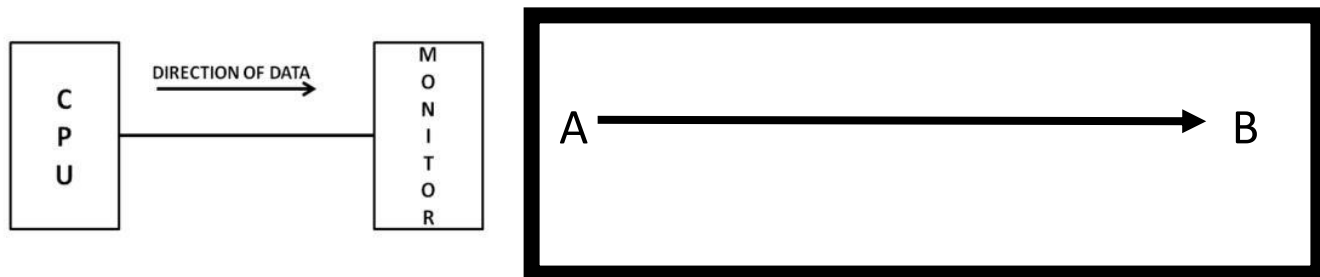
Analog	Digital
A continuous wave that represents data.	A discrete signal that represents data in binary form (0s and 1s).
Requires less bandwidth for transmission.	Requires more bandwidth due to discrete data encoding.
Less expensive to produce and transmit.	More expensive due to advanced technology and processing requirements.
Less flexible. harder to modify or enhance.	Highly flexible. easy to modify, compress, or enhance.

❖ Types of communication channels

1. Simplex Channel

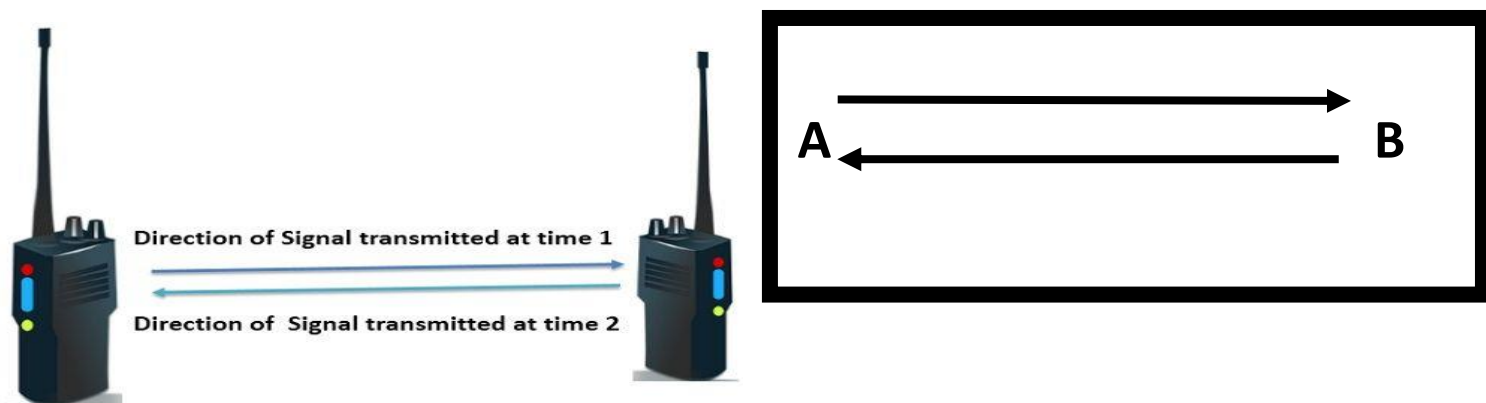
- In this, the flow of data is always in one direction.
- This channel can transmit data in only one direction.

- Through this channel, only one transmitting device can send information and the other transmitting device can only receive information.
- For example- 1. Radio station from reach the listeners, but do not go back from the listeners to the radio station. 2. CPU to Monitor-A cpu send data while a monitor only receives data.



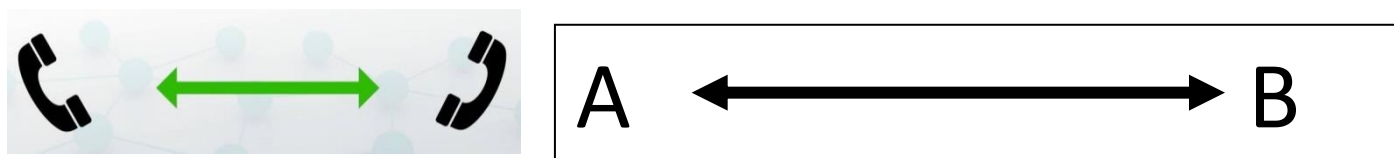
2. Half Duplex Channel

- In this channel, data flows in both directions, but data can flow in only one direction at a time.
- When one device is sending other can only receive and viceversa
- For example, in old telephone line, data is transmitted in only one direction at a time, A walkie-talkie.



3. Full Duplex Channel

- In Full duplex mode, flows data on both side (Sender & Reciver) can transmit and receive at the same time.
- Example: mobile phones



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